Physiology of Blood & Immunity: Set 2

1. During the biochemical events of local tissue inflammation, why do circulating neutrophils selectively accumulate and bind to the endothelial walls of the capillaries near the injury site rather than floating past?
a. Because systemic arterial blood pressure drops to zero locally
b. Because activated tissue macrophages release lysosomal enzymes that dissolve the plasma albumin
c. Because endothelial surface tight junctions completely disappear
d. Because local inflammatory cytokines induce the expression of selectins and integrins (cell adhesion molecules) on both endothelial cells and neutrophils
e. Because the neutrophils release a dense layer of heparin matrix
2. A tissue macrophage can ingest much larger foreign particles, including entire red blood cells and necrotic tissue debris, compared to a circulating neutrophil. This superior phagocytic capacity is conceptually due to the macrophage possessing:
a. A lack of an intracellular Golgi apparatus
b. A highly electronegative cell wall membrane
c. An exceptionally large volume of cytoplasm packed with lysosomes and a larger cell size
d. Continuous active synthesis of hemoglobin alpha chains
e. Zero surface Major Histocompatibility Complex proteins
3. Conceptually, how do professional opsonins like IgG antibodies or complement fragment C3b physically accelerate the rate of phagocytosis when they coat an invading bacterium?
a. They bind directly to specific surface receptors on the phagocyte membrane, anchoring the invader and triggering immediate pseudopod evagination
b. They alter the bacterial genetic material to turn off cell wall lipids
c. They synthesize a layer of yellow elastic cartilage around the bacterium
d. They convert the soluble plasma fibrinogen into an insoluble fibrin mesh
e. They activate the myenteric Auerbach's plexus to drive peristalsis
4. When a neutrophil executes its bactericidal function within an inflammatory zone, it undergoes an explosive metabolic event known as the respiratory burst. The fundamental physiological purpose of this burst is to:
a. Accelerate the rate of mitochondrial oxidative phosphorylation to synthesize ATP
b. Increase the intracellular concentration of molecular oxygen to produce lethal reactive oxygen species (ROS) like superoxide and hydroxyl radicals
c. Drive the extrusion of the condensed cell nucleus
d. Stimulate the liver to release massive quantities of transferrin
e. Intravascularly activate the intrinsic coagulation pathway
5. Conceptually, what is the core mechanism that prevents the massive, destructive enzymes contained within a leukocyte's lysosomes from digesting and killing the host white blood cell before it encounters a foreign pathogen?
a. The protective enzymes are kept locked inside a highly resilient, specialized lysosomal membrane that keeps them sequestered from the cytoplasm
b. The internal cytoplasm of a leukocyte features an exceptionally high alkaline pH that punches out acids
c. Host leukocytes completely lack any internal intracellular proteins or structural lipids
d. The plasma albumin acts as a continuous buffer inside the cell wall

e. Lysosomal enzymes only fold into an active shape when they touch bile salts

6. During the progression of localized tissue inflammation, the tissue macrophage system acts as the absolute first line of defense. However, within a few hours, the neutrophil invasion takes over. The physiological reason for this chronological shift is that:

a. Neutrophils undergo rapid mitosis inside the local inflamed tissue

b. Circulating neutrophils are already fully mature, exist in vast quantities in the blood, and respond rapidly to local chemotactic gradients

c. Monocytes actively destroy all tissue macrophages to clear space

d. The kidney ceases the production of erythropoietin during the acute hour

e. Mast cells systematically convert all erythrocytes into granulocytes

7. In an individual experiencing a severe chronic parasitic infection (such as trichinosis or schistosomiasis), a marked systemic eosinophilia is observed. Eosinophils achieve their specific larvicidal action primarily by:

a. Ingesting the entire multicellular parasite via phagocytosis

b. Releasing massive quantities of histamine to paralyze the worm muscle bundles

c. Attaching to the parasite surface and releasing lethal proteins, such as major basic protein and reactive oxygen species, directly onto the invader

d. Activating the membrane attack complex of the clotting cascade

e. Shifting the oxygen-hemoglobin dissociation curve to the left

8. Conceptually, how does the physiological role of tissue mast cells and circulating basophils alter the structural properties of local microvessels during an immediate hypersensitivity (allergic) reaction?

a. Released histamine causes intense vasoconstriction of large systemic arteries

b. Released histamine binds to endothelial receptors, driving local arteriolar vasodilation and a massive increase in capillary permeability

c. They release hepcidin to completely stop the filtration of fluid

d. They deposit calcified acervuli grains into the vessel adventitia

e. They consume all circulating IgE molecules to cause vascular spasm

9. Innate immunity relies on immediate, non-specific mechanisms to protect the host. In contrast, acquired immunity demonstrates extreme antigen specificity and immunological memory. This long-term adaptation is conceptually achieved because acquired immunity utilizes:

a. Non-specific chemical agents like lysozymes and basic polypeptides

b. The continuous active phagocytic tracking of neutrophils

c. The secretion of low-pressure hydrochloric acid by parietal cells

d. The structural rigidity of the C-shaped tracheal cartilage rings

e. Proliferating clones of antigen-specific T and B lymphocytes that form persistent memory cells

10. B lymphocytes undergo processing in the bone marrow, while T lymphocytes undergo processing in the thymus. The fundamental physiological consequence of this processing phase is to:

a. Induce the cells to synthesize large stores of hemoglobin HbF

b. Ensure that both lymphocyte lineages lose their entire nuclear genomes

c. Generate millions of distinct clones, each recognizing a specific antigen, while deleting clones that react against the body's own self-antigens

d. Stimulate the direct extracellular assembly of the clotting cascade Stage 2

e. Transform the lymphocytes into non-nucleated platelet fragments

11. The structural diversity of the variable regions (antigen-binding sites) of an antibody molecule is immense, allowing the body to bind to millions of distinct foreign epitopes. This structural variability is genetically generated by:
a. Continuous random point mutations in the liver cells after birth
b. Somatic recombination and shuffling of Variable (V), Diversity (D), and Joining (J) gene segments during lymphocyte development
c. The rate of dietary iron absorption in the intestinal tract duodenum
d. Shifting the concentration of plasma albumin from alpha to beta globulins
e. Direct enzymatic translation of the core cardiac conduction paths
12. When a specific B lymphocyte is activated by its corresponding antigen, it undergoes clonal expansion and differentiates into a population of large, protein-synthesizing factory cells known as:
a. Helper T cells (CD4)
b. Cytotoxic T cells (CD8)
c. Plasma cells (which actively secrete thousands of antibody molecules per second)
d. Alveolar macrophages
e. Type II pneumocytes
13. Conceptually, how does the structural mechanism of an antibody monomer change if it belongs to the IgM class rather than the IgG class?
a. The IgM antibody completely lacks variable antigen-binding sites
b. The IgM antibody features a single heavy chain bound to a heme atom
c. Five monomeric antibody units are cross-linked together by a joining (J) chain to form a pentamer with 10 operational binding sites
d. It is completely lipid-soluble to track through the brain barrier
e. It possesses a flat, biconcave profile optimized for gas diffusion
14. Antibodies neutralize toxins or agglutinate bacteria directly. However, the most destructive, high-yield action of the humoral immune response occurs when the antibody's Fc portion binds to and activates the C1 component of the complement system, which initiates:
a. The negative feedback inhibition of erythropoietin
b. An enzymatic cascade that generates anaphylatoxins, opsonins, and punches structural holes through the invader via the membrane attack complex
c. The immediate beta-oxidation of long association white matter tracks
d. The cross-linking of the loose inner circular muscularis externa layers
e. A massive reduction in the rate-limiting step of prothrombin synthesis
15. The complement cascade generates a cleavage fragment known as C3b. The major high-yield conceptual role of C3b within an active immune zone is to function as a highly powerful:
a. Local vasodilator autacoid
b. Involuntary smooth muscle contractable protein
c. Specific antibody variable region look-alike
d. Opsonin (coating the pathogen to strongly stimulate macrophage phagocytosis)
e. Primary pacemaking node for cellular replication
16. Cell-mediated immunity relies on T lymphocytes recognizing presented antigens. Conceptually, why can a T-cell receptor (TCR) *never* bind to a naked, free-floating viral particle circulating inside the blood plasma?
a. Because T cells completely lack any surface glycoprotein receptors

b. Because T-cell receptors can only recognize and bind to processed peptide antigens that are physically anchored within host Major Histocompatibility Complex (MHC) proteins on an antigen-presenting cell surface
c. Because free viruses always carry a massive electronegative charge
d. Because circulating plasma albumin wraps all viruses in a tight shell
e. Because the T cell requires a direct synapse with the optic chiasm

17. Class I MHC proteins present antigens to one specific lineage of T lymphocytes, while Class II MHC proteins present to another. Class II MHC proteins are expressed exclusively on the surfaces of:

a. Every single nucleated cell throughout the human body
b. Non-nucleated mature human erythrocytes
c. Professional Antigen-Presenting Cells (APCs, including macrophages, dendritic cells, and B lymphocytes)
d. The myelin sheath tracks synthesized by oligodendrocytes
e. Intrinsic laryngeal muscles derived from the sixth arch

18. Helper T cells (CD4+ T cells) represent the absolute master regulators of the entire acquired immune system. They execute this coordinated control primarily by synthesizing and secreting signaling molecules called:

a. Antibodies (immunoglobulins)
b. Lymphokines or cytokines (such as IL-2, IL-3, IL-4, IL-5, and GM-CSF)
c. Proteolytic lysosomal enzymes like myeloperoxidase
d. Local autacoids like thromboxane A2 and serotonin
e. Carbonic anhydrase polymers

19. In an individual who has been infected with the Human Immunodeficiency Virus (HIV), the virus selectively targets and destroys CD4+ T cells. This leads to a total collapse of both humoral and cell-mediated immunity because a lack of helper T-cell lymphokines prevents:

a. The bone marrow from synthesizing pluripotent stem cells initially
b. The activation, proliferation, and differentiation of antigen-specific B cells and cytotoxic T cells
c. The direct physical migration of erythrocytes through capillary pores
d. The conjugation of free bilirubin with glucuronic acid inside the liver
e. The relaxation of the posterior trachealis smooth muscle band

20. Cytotoxic T cells (CD8+ T cells) specialize in destroying virus-infected host cells and tumor cells. They achieve this targeted killing by releasing specialized proteins that punch holes in the target membrane and induce apoptosis, namely:

a. Histamine and holiday autacoids
b. Perforins (hole-forming proteins) and granzymes (proteolytic enzymes)
c. Heparin and antithrombin III complexes
d. Major basic protein and eosinophil cationic factor
e. Fibrinogen monomers and prothrombin activator units

21. Suppressor T cells (Regulatory T cells / Tregs) play a vital physiological role in maintaining immune homeostasis. Their primary functional concept is to:

a. Suppress the actions of cytotoxic and helper T cells, preventing excessive, life-threatening autoimmune destruction of host tissues
b. Move the rate of antibody secretion by active plasma cells upward
c. Drive the chemotaxis of neutrophils to a localized inflammatory zone
d. Upregulate the expression of Class II MHC proteins on all cell surfaces

e. Stimulate the rapid fragmentation of megakaryocytes into platelets

22. A primary immune response occurs upon first exposure to an antigen, displaying a long lag phase and producing low antibody titers. A secondary immune response occurs upon subsequent re-exposure, showing a rapid, explosive elevation in antibody levels due to the presence of:

a. Highly active tissue macrophages already coating the area

b. Large pools of specific, long-lived Memory B and Memory T cells created during the primary response

c. High baseline levels of circulating erythropoietin hormone

d. The smooth glycocalyx repelling forces on endothelial walls

e. Insoluble cross-linked fibrin polymers inside the marrow stroma

23. Conceptually, what is the core immunological defect that drives an immediate Type I hypersensitivity (anaphylactic) reaction when an atopic individual is exposed to an environmental allergen?

a. Widespread immune complexes deposit inside the basement membrane of systemic blood vessels

b. Cytotoxic T cells directly attack and destroy healthy skin epithelial cells

c. The individual's immune system manufactures large amounts of allergen-specific IgE antibodies, which bind tightly to the surface of mast cells and basophils, triggering explosive degranulation upon re-exposure

d. The complement system fails to construct the membrane attack complex

e. The liver ceases the synthesis of all plasma clotting factors

24. In a patient experiencing systemic anaphylactic shock following a penicillin injection, a widespread, massive release of histamine into the circulation causes a life-threatening emergency characterized by:

a. Severe hypertension and a total loss of capillary permeability

b. Widespread arteriolar vasodilation and extreme loss of plasma fluid into the tissues, causing a critical drop in blood pressure and severe bronchoconstriction

c. Sudden hypercalcemia and muscle rigidity

d. Sluggish blood flow due to an immediate rise in the hematocrit level

e. Localized calcification of the pineal gland brain sand

25. Type II hypersclerosis reactions are cytotoxic reactions. Conceptually, this reaction occurs when antibodies (IgG or IgM) bind directly to antigens on the surface of specific cells, leading to target destruction via:

a. The release of major basic protein from activated eosinophils

b. Localized ischemic necrosis driven by a tortuous splenic artery

c. Complement-mediated lysis or opsonization and phagocytosis (such as in a mismatched blood transfusion reaction)

d. The slow, vertical downward growth of the embryonic septum primum

e. The selective degradation of Factors Va and VIIIa by Protein C

26. Type III hypersensitivity reactions are immune complex-mediated disorders. This pathology occurs conceptually when large numbers of soluble antigen-antibody complexes form in the blood and become pathologically:

a. Excreted by the kidneys into the urine matrix

b. Incorporated into the cytoplasm of developing proerythroblasts

c. Transformed into active prothrombin activator complexes

d. Dissolved by tissue plasminogen activator (tPA)

e. Trapped and deposited within the endothelial lining of blood vessels or tissues, triggering complement activation and severe neutrophilic inflammation

27. Type IV hypersensitivity differs fundamentally from Types I, II, and III because it is completely independent of antibodies. Type IV (delayed-type) hypersensitivity is mediated entirely by:

a. Basophils releasing histamine from delta granules
b. Activated B lymphocytes expanding into plasma cells
c. Activated T lymphocytes (helper and cytotoxic T cells) and macrophages, typically reaching peak inflammation 24 to 72 hours after exposure
d. The rapid intravascular deposition of insoluble fibrin mesh
e. The sudden contraction of the smooth trachealis muscle line

28. Conceptually, why does the physiological layout of the human lymphatic system ensure that all lymph fluid collected from peripheral tissue spaces must pass through at least one lymph node before emptying into the venous system?

a. To maximize the concentration of plasma albumin in the lymph
b. To expose the fluid systematically to thousands of localized, highly active macrophages, B cells, and T cells to filter out and destroy any entering pathogens
c. To completely remove all water molecules from the tissue spaces
d. To accelerate the rate-limiting step of reticulocyte maturation
e. To fix the lymph fluid securely to the cricoid cartilage table

29. Natural Killer (NK) cells form a key component of the innate immune defense. Unlike acquired cytotoxic T cells, NK cells are capable of identifying and killing virus-infected or malignant host cells without:

a. Utilizing any cytoplasmic granules or perforin proteins
b. Possessing an internal cell nucleus or mitochondria
c. Prior specific antigen sensitization or matching Major Histocompatibility Complex (MHC) presentation requirements
d. Circulating within the vascular endothelial channels
e. Activating the complement cascade system

30. The phenomenon of immunological tolerance ensures that the acquired immune system does not attack the body's own tissues. Tolerance is primarily established through clonal deletion and clonal anergy during development, processing centers that operate inside the:

a. Spleen and liver hila
b. Bone marrow and thymus gland (central tolerance processing sites)
c. Palatine tonsil crypts and retropharyngeal spaces
d. Main pancreatic duct and major duodenal papilla
e. Rhomboid fossa and fourth ventricle floor